

Impinj R700 RAIN RFID Reader

The Impinj R700 is a high-performance four-port fixed RAIN RFID reader designed for enterprise-grade deployments and next-generation IoT solutions. Powered by a dual-core 1 GHz ARM processor with 1 GB RAM, native MQTT support, and an upgradable Linux OS, it delivers industry-leading read sensitivity and edge processing.



RFID

Air Protocol	EPCglobal UHF Class 1 Gen 2 / ISO 18000-63
Transmit Power	10 – 30 dBm (PoE); 10 – 31.5 dBm (PoE+, ETSI Lower); 10 – 33 dBm (PoE+, FCC / ETSI Upper)
Transmit Power Resolution	0.25 dB
Transmit Power Accuracy	+/- 0.5 dB
Frequency Range	IPJ-R700-241 ETSI: 865 – 868 MHz, 915 – 921 MHz
Return Loss	10 dB min
Read Rate	Up to 1100 reads per second
Antenna Impedance	50 ohms
Max Receive Sensitivity	-92 dBm

MECHANICAL

Physical Dimensions	With mounting brackets 21.5 x 18.7 x 3.0 cm; Without brackets 21.4 x 14.9 x 3.0 cm
Mounting	VESA 100 × 100 mm; Two removable Speedway-compatible brackets
Weight	2.12 lb / 0.96 kg
Housing Material	Die-cast Aluminum
LED Indicators	System status, inventory, firmware upgrade, antenna activity, network activity

CONNECTIVITY

Network	10/100/1000 Base-T Ethernet (RJ45)
Antenna Ports	4 monostatic ports (RP-TNC)
USB	3 Type A host, 1 micro device
Power	PoE (802.3af), PoE+ (802.3at) with LLDP
General Purpose I/O	3 out, 2 in, optically isolated; Phoenix Contact 9-pin 3.81 mm header

ENVIRONMENTAL

Operating Temperature	-20 °C to 50 °C
Storage Temperature	-20 °C to 70 °C
Humidity	5% - 95% non-condensing
Sealing	IP50
Shock and Vibration	MIL-STD-810G

OPERATING SYSTEM

Processor	Dual-Core 1 GHz Cortex A7
Memory	1 GB Flash, 1 GB RAM
Operating System	Linux, 5.4 kernel
Network Services	SSH, HTTP, HTTPS, NTP, DHCP, SFTP, mDNS
Network Security	802.1x port security

DEVELOPMENT LIBRARIES

Reader configuration	OpenAPI compatible RESTful API
Impinj Octane SDK	.NET, Java
Impinj Octane LLRP toolkit	.NET, Java, C, C++
Impinj R700 embedded toolkit	C, C++ compiler with precompiled Octane LLRP libraries

REGULATORY

USA and Canada	FCC Part 15.247 and 15.249; UL listed
EU	EN 302 208 v3.3.1; EN 301 489-3 v2.2.1; EN 300 489-1 v2.2.0; CE mark
Japan	ARIB STD-T106
Hazardous Substances	RoHS3 2015/863 compliant

FEATURE HIGHLIGHTS

- Industry-leading sensitivity for reading tags farther and faster
- Powerful edge processing with intelligent on-reader algorithms
- Secure, upgradable Linux OS for enterprise-grade reliability
- Simple IoT device interface with native MQTT support
- Optimized design for inventory with higher read rate at lower power
- Rich peripheral and accessory support (USB, Wi-Fi adapters, up to 32 antennas)

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed RAIN RFID Reader